

EE114 Homework 3

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Responses

1. Question 2.2.2

Question: For random variables X and R defined in Example 2.5, find $P_X(x)$ and $P_R(r)$. In addition, find the following probabilities:

- (a) $P[X = 0]$
- (b) $P[X < 3]$
- (c) $P[R > 1]$

Response:

Given three calls, each is voice (v) or data (d), equally likely. Total outcomes = 8, each with probability $1/8$.

Let X be the number of voice calls, $Y = 3 - X$, and $R = XY$.

PMF of X :

x	0	1	2	3
$P_X(x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

PMF of R :

Compute $R = X(3 - X)$:

- $R = 0$ when $X = 0$ or $X = 3 \Rightarrow P[R = 0] = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$.
- $R = 2$ when $X = 1$ or $X = 2 \Rightarrow P[R = 2] = \frac{3}{8} + \frac{3}{8} = \frac{3}{4}$.

$$P_R(r) = \begin{cases} \frac{1}{4}, & r = 0, \\ \frac{3}{4}, & r = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Probabilities:

(a) $P[X = 0] = \frac{1}{8}.$

(b) $P[X < 3] = 1 - P[X = 3] = 1 - \frac{1}{8} = \frac{7}{8}.$

(c) $P[R > 1] = P[R = 2] = \frac{3}{4}.$

2. Question 2.2.4

Question: The random variable X has PMF

$$P_X(x) = \begin{cases} \frac{c}{x}, & x \in \{2, 4, 8\}, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[X = 4]$?
- (c) What is $P[X < 4]$?
- (d) What is $P[3 \leq X \leq 9]$?

Response:

Given

$$P_X(x) = \begin{cases} \frac{c}{x}, & x \in \{2, 4, 8\}, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Since the PMF must sum to 1,

$$\frac{c}{2} + \frac{c}{4} + \frac{c}{8} = 1 \quad \Rightarrow \quad c \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) = c \left(\frac{7}{8} \right) = 1 \quad \Rightarrow \quad c = \frac{8}{7}.$$

- (b)

$$P[X = 4] = \frac{c}{4} = \frac{8/7}{4} = \frac{2}{7}.$$

- (c) The only value less than 4 is $X = 2$, so

$$P[X < 4] = P[X = 2] = \frac{c}{2} = \frac{8/7}{2} = \frac{4}{7}.$$

- (d) Values in the range $3 \leq X \leq 9$ are 4 and 8, so

$$P[3 \leq X \leq 9] = P[X = 4] + P[X = 8] = \frac{2}{7} + \frac{1}{7} = \frac{3}{7}.$$

3. Question 2.2.7

Question: In the ticket agency of Problem 2.2.6, each telephone ticket agent is available to receive a call with probability 0.2. If all agents are busy when someone calls, the caller hears a busy signal. What is the minimum number of agents that you have to hire to meet your goal of serving 95% of the customers who want tickets?

Response:

A caller tries 3 times. Let p be the probability a call gets through.

Probability a caller is never served: $(1 - p)^3$.

To serve at least 95% of callers:

$$1 - (1 - p)^3 \geq 0.95 \quad \Rightarrow \quad (1 - p)^3 \leq 0.05 \quad \Rightarrow \quad 1 - p \leq 0.05^{1/3} \approx 0.368 \quad \Rightarrow \quad p \geq 0.632.$$

Relate p to number of agents: Each agent is available with probability 0.2, independently. If there are N agents, the probability all are busy is $(0.8)^N$, so

$$p = 1 - (0.8)^N.$$

We need

$$1 - (0.8)^N \geq 0.632 \quad \Rightarrow \quad (0.8)^N \leq 0.368.$$

Test powers of 0.8:

$$0.8^4 = 0.4096 \text{ (too large),} \quad 0.8^5 = 0.32768 \text{ (works).}$$

Final answer: $N = 5$ agents.

4. Question 2.3.2

Question: In the ticket agency of Problem 2.2.6, each telephone ticket agent is available to receive a call with probability 0.2. If all agents are busy when someone calls, the caller hears a busy signal. What is the minimum number of agents that you have to hire to meet your goal of serving 95% of the customers who want tickets?

Response:

Step 1: Required probability of getting through (from Problem 2.2.6).

A caller fails all 3 attempts with probability $(1 - p)^3$. Require

$$1 - (1 - p)^3 \geq 0.95 \quad \Rightarrow \quad (1 - p)^3 \leq 0.05 \quad \Rightarrow \quad p \geq 0.632.$$

Step 2: Express p in terms of number of agents N .

Agent busy probability = 0.8, so all N agents busy is $(0.8)^N$. Hence

$$p = 1 - (0.8)^N.$$

Require

$$1 - (0.8)^N \geq 0.632 \quad \Rightarrow \quad (0.8)^N \leq 0.368.$$

Step 3: Find minimum N .

$$0.8^4 = 0.4096 \text{ (too large),} \quad 0.8^5 \approx 0.3277 \text{ (works).}$$

Final Answer: $N = 5$ agents.

5. Question 2.3.6

Question: The number of bits B in a fax transmission is a geometric ($p = 2.5 \cdot 10^{-5}$) random variable. What is the probability $P[B > 500,000]$ that a fax has over 500,000 bits?

Response:

Step 1: Required probability of getting through (from Problem 2.2.6).

A caller fails all 3 attempts with probability $(1 - p)^3$. Require

$$1 - (1 - p)^3 \geq 0.95 \quad \Rightarrow \quad (1 - p)^3 \leq 0.05 \quad \Rightarrow \quad p \geq 0.632.$$

Step 2: Express p in terms of number of agents N .

Agent busy probability = 0.8, so all N agents busy is $(0.8)^N$. Hence

$$p = 1 - (0.8)^N.$$

Require

$$1 - (0.8)^N \geq 0.632 \quad \Rightarrow \quad (0.8)^N \leq 0.368.$$

Step 3: Find minimum N .

$$0.8^4 = 0.4096 \text{ (too large)}, \quad 0.8^5 \approx 0.3277 \text{ (works)}.$$

Final Answer: $N = 5$ agents.

6. Question 2.3.7

Question: The number of buses that arrive at a bus stop in T minutes is a Poisson random variable B with expected value $T/5$.

- (a) What is the PMF of B , the number of buses that arrive in T minutes?
- (b) What is the probability that in a two-minute interval, three buses will arrive?
- (c) What is the probability of no buses arriving in a 10-minute interval?
- (d) How much time should you allow so that with probability 0.99 at least one bus arrives?

Response:

Let $B \sim \text{Poisson}(\lambda)$ with $\lambda = \frac{T}{5}$.

- (a) For $k = 0, 1, 2, \dots$,

$$P(B = k) = \frac{\lambda^k e^{-\lambda}}{k!} = \frac{\left(\frac{T}{5}\right)^k e^{-T/5}}{k!}.$$

- (b) For $T = 2$, $\lambda = \frac{2}{5} = 0.4$:

$$P(B = 3) = \frac{(0.4)^3 e^{-0.4}}{3!}.$$

- (c) For $T = 10$, $\lambda = \frac{10}{5} = 2$:

$$P(B = 0) = e^{-2}.$$

- (d) Want $P(B \geq 1) = 0.99$.

$$P(B \geq 1) = 1 - P(B = 0) = 1 - e^{-T/5} = 0.99 \Rightarrow e^{-T/5} = 0.01 \Rightarrow T = -5 \ln(0.01) \approx 23 \text{ min.}$$

7. Question 2.3.8

Question: In a wireless automatic meter reading system, a base station sends out a wake-up signal to nearby electric meters. On hearing the wake-up signal, a meter transmits a message indicating the electric usage. Each message is repeated eight times.

- (a) If a single transmission of a message is successful with probability p , what is the PMF of N , the number of successful message transmissions?
- (b) I is an indicator random variable such that $I = 1$ if at least one message is transmitted successfully; otherwise $I = 0$. Find the PMF of I .

Response:

- (a) The number of successful transmissions N is binomial:

$$N \sim \text{Binomial}(8, p).$$

Hence, for $k = 0, 1, \dots, 8$,

$$P(N = k) = \binom{8}{k} p^k (1 - p)^{8-k}.$$

- (b) Let $I = 1$ if at least one transmission succeeds, and $I = 0$ otherwise.

$$P(I = 1) = 1 - P(\text{no successes}) = 1 - (1 - p)^8, \quad P(I = 0) = (1 - p)^8.$$

$$P_I(i) = \begin{cases} 1 - (1 - p)^8, & i = 1, \\ (1 - p)^8, & i = 0, \\ 0, & \text{otherwise.} \end{cases}$$

8. Question 2.3.11

Question: In a packet voice communications system, a source transmits packets containing digitized speech to a receiver. Because transmission errors occasionally occur, an acknowledgment (ACK) or a nonacknowledgment (NAK) is transmitted back to the source to indicate the status of each received packet. When the transmitter gets a NAK, the packet is retransmitted. Voice packets are delay sensitive and a packet can be transmitted a maximum of d times. If a packet transmission is an independent Bernoulli trial with success probability p , what is the PMF of T , the number of times a packet is transmitted?

Response:

Case 1: Packet succeeds on attempt t , where $t = 1, 2, \dots, d - 1$.

To stop at time t :

- First $t - 1$ attempts fail.
- The t -th attempt succeeds.

$$P(T = t) = (1 - p)^{t-1}p, \quad t = 1, 2, \dots, d - 1.$$

Case 2: Packet is transmitted d times.

This happens if all first $d - 1$ attempts fail (regardless of success on the d -th attempt, since retransmissions stop at d):

$$P(T = d) = (1 - p)^{d-1}.$$

Final PMF:

$$P(T = t) = \begin{cases} (1 - p)^{t-1}p, & t = 1, 2, \dots, d - 1, \\ (1 - p)^{d-1}, & t = d, \\ 0, & \text{otherwise.} \end{cases}$$

9. Question 2.4.2

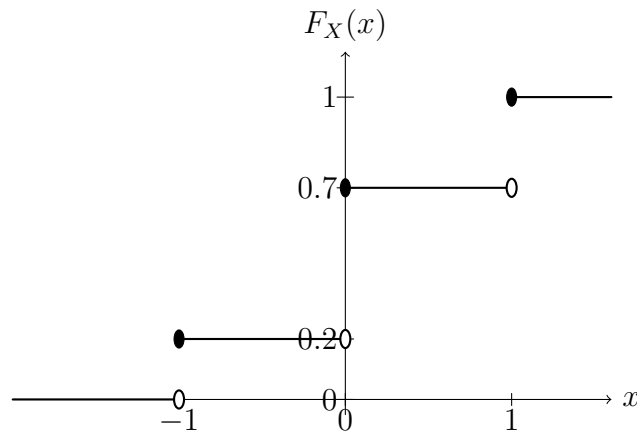
Question: The random variable X has CDF

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \leq x < 0, \\ 0.7, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

- (a) Draw a graph of the CDF.
- (b) Write $P_X(x)$, the PMF of X . Be sure to write the value of $P_X(x)$ for all x from $-\infty$ to ∞ .

Response:

- (a) **Graph of the CDF:**



- (b) **PMF $P_X(x)$:** The PMF is the size of each jump in the CDF.

$$P_X(x) = \begin{cases} 0.2, & x = -1, \\ 0.5, & x = 0, \\ 0.3, & x = 1, \\ 0, & \text{otherwise.} \end{cases}$$

Check: $0.2 + 0.5 + 0.3 = 1$.

10. Question 2.4.4

Question: Following Example 2.24, show that a geometric (p) random variable K has CDF

$$F_K(k) = \begin{cases} 0, & k < 1, \\ 1 - (1 - p)^{\lfloor k \rfloor}, & k \geq 1. \end{cases}$$

Response:

Let $K \sim \text{Geometric}(p)$, where K is the number of trials until the first success. Then

$$P(K = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$$

By definition, the CDF is $F_K(k) = P(K \leq k)$.

- **Case 1:** $k < 1$. Since $K \geq 1$ always,

$$F_K(k) = 0, \quad k < 1.$$

- **Case 2:** $k \geq 1$.

$$F_K(k) = \sum_{i=1}^{\lfloor k \rfloor} (1 - p)^{i-1}p = p \sum_{i=0}^{\lfloor k \rfloor - 1} (1 - p)^i.$$

This is a geometric series, so

$$F_K(k) = p \cdot \frac{1 - (1 - p)^{\lfloor k \rfloor}}{1 - (1 - p)} = 1 - (1 - p)^{\lfloor k \rfloor}, \quad k \geq 1.$$

Final CDF:

$$F_K(k) = \begin{cases} 0, & k < 1, \\ 1 - (1 - p)^{\lfloor k \rfloor}, & k \geq 1. \end{cases}$$